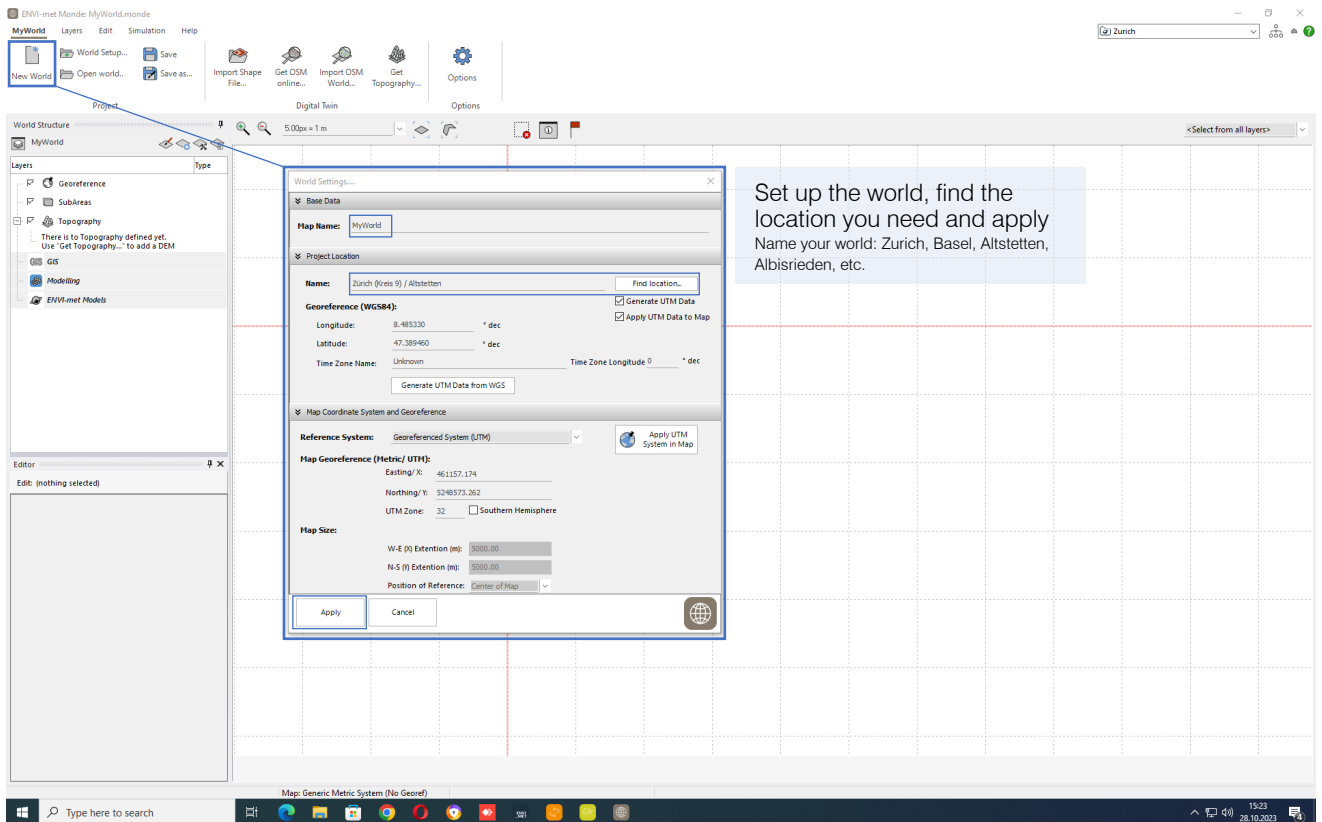
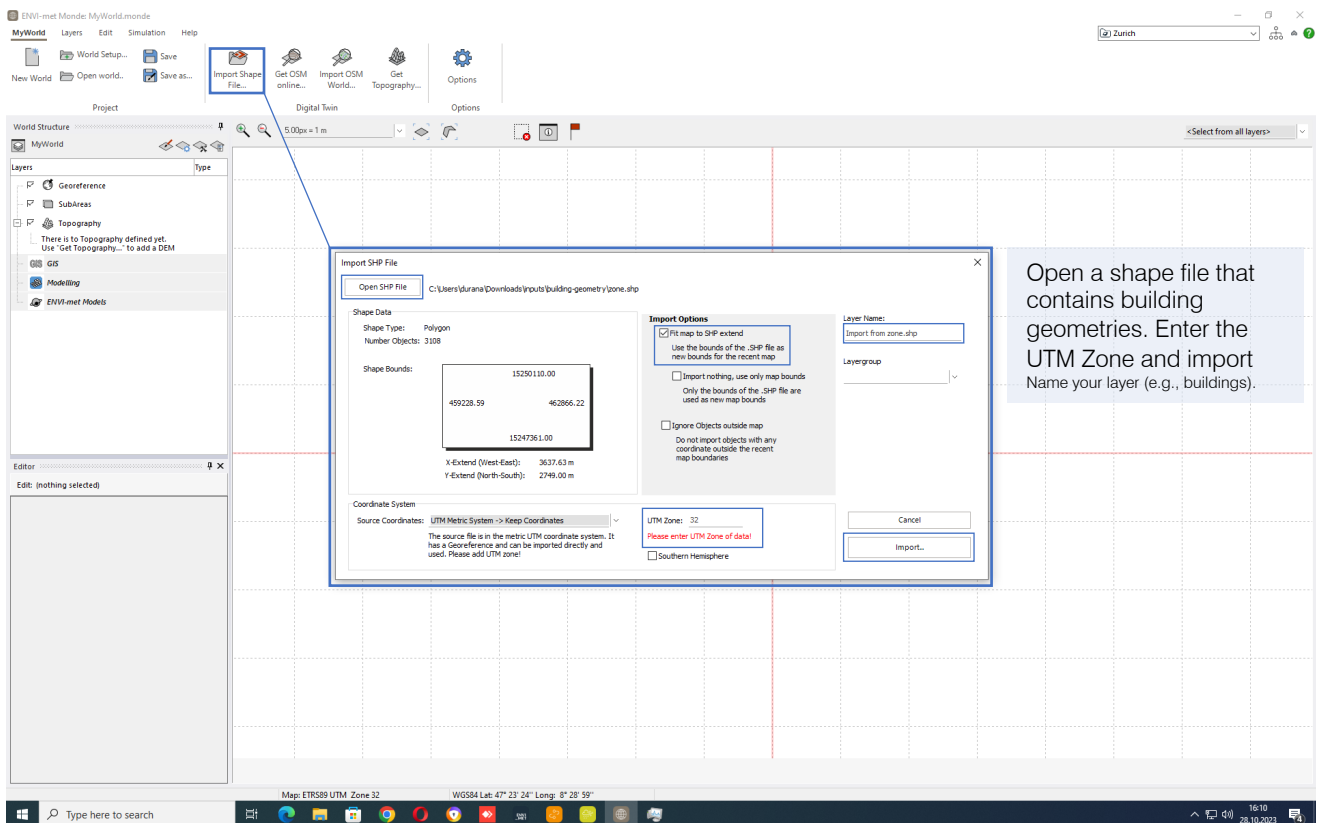


Modeling with Monde

1

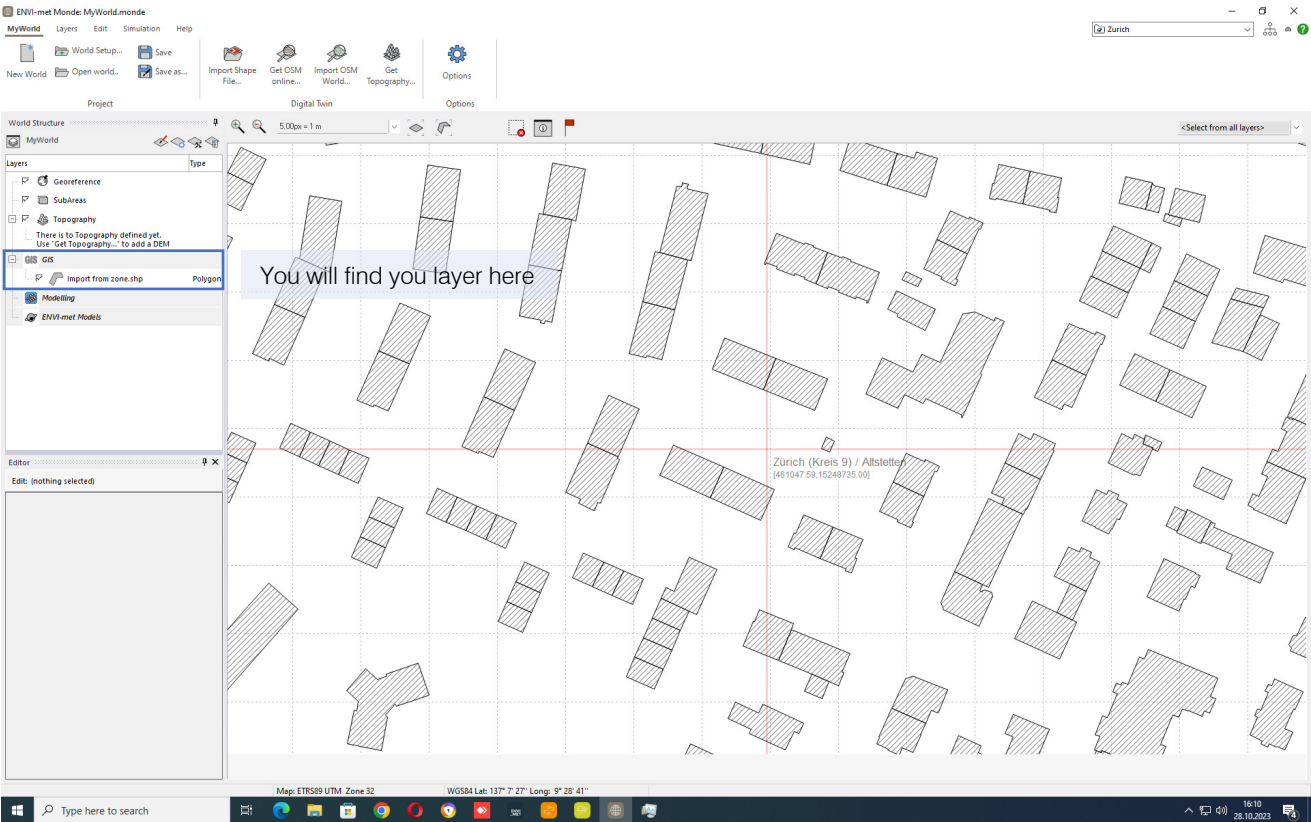


2

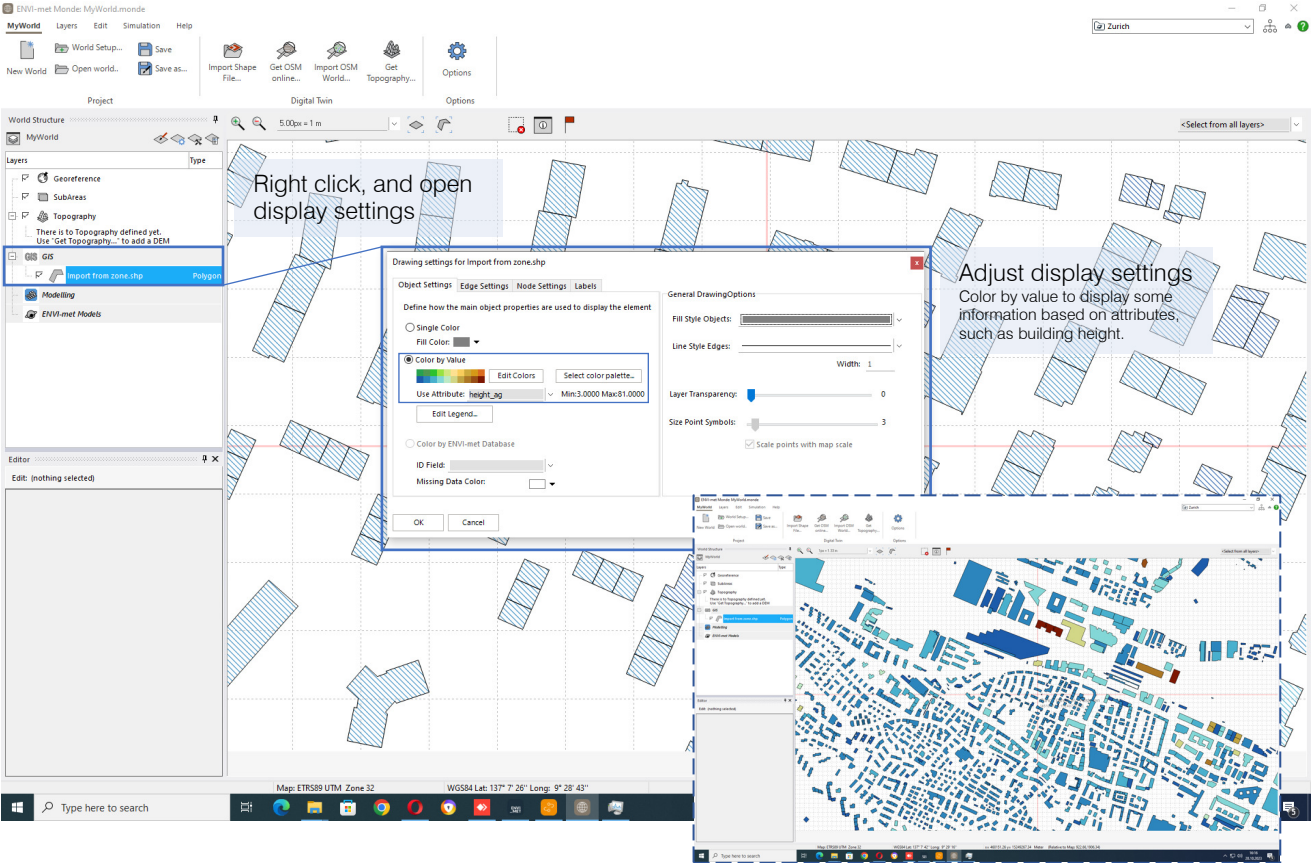


Modeling

3

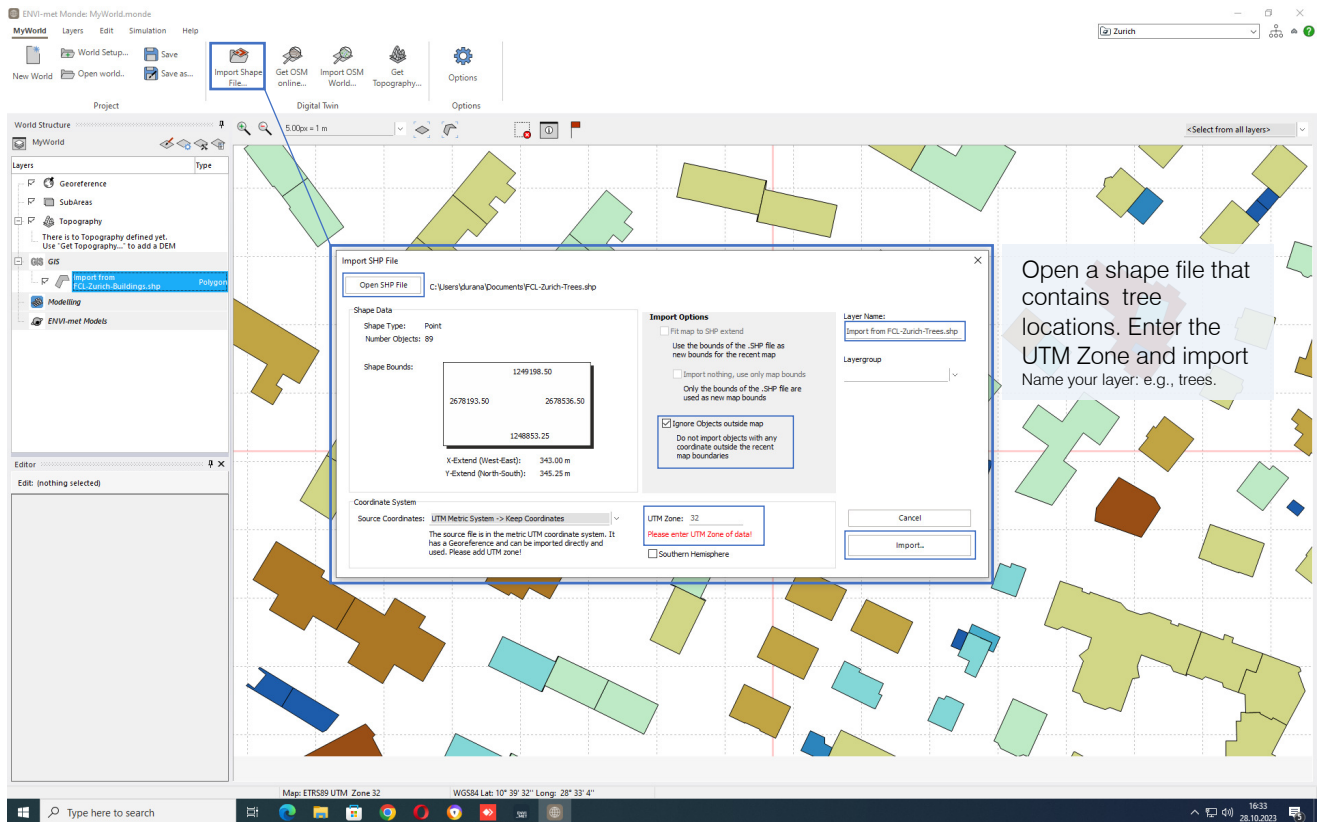


4

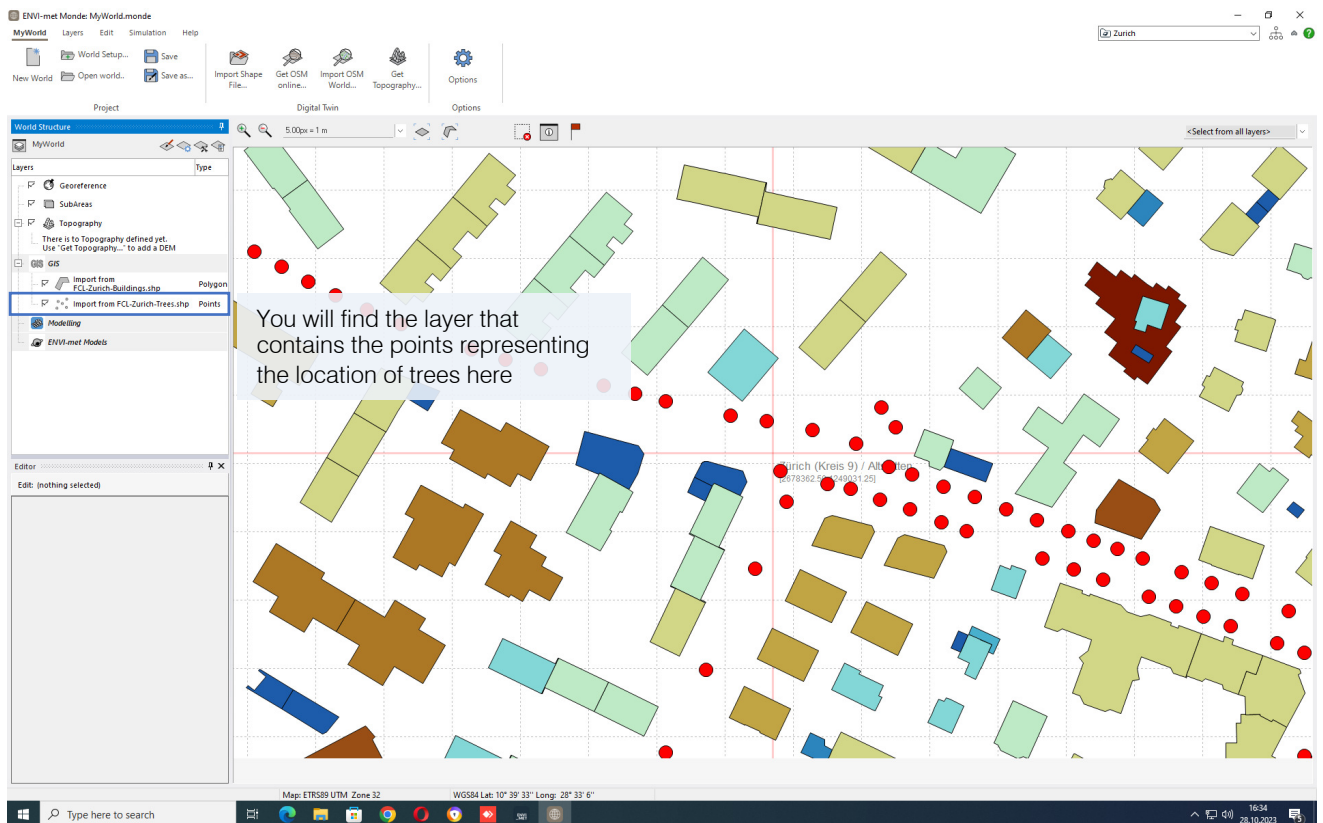


Modeling

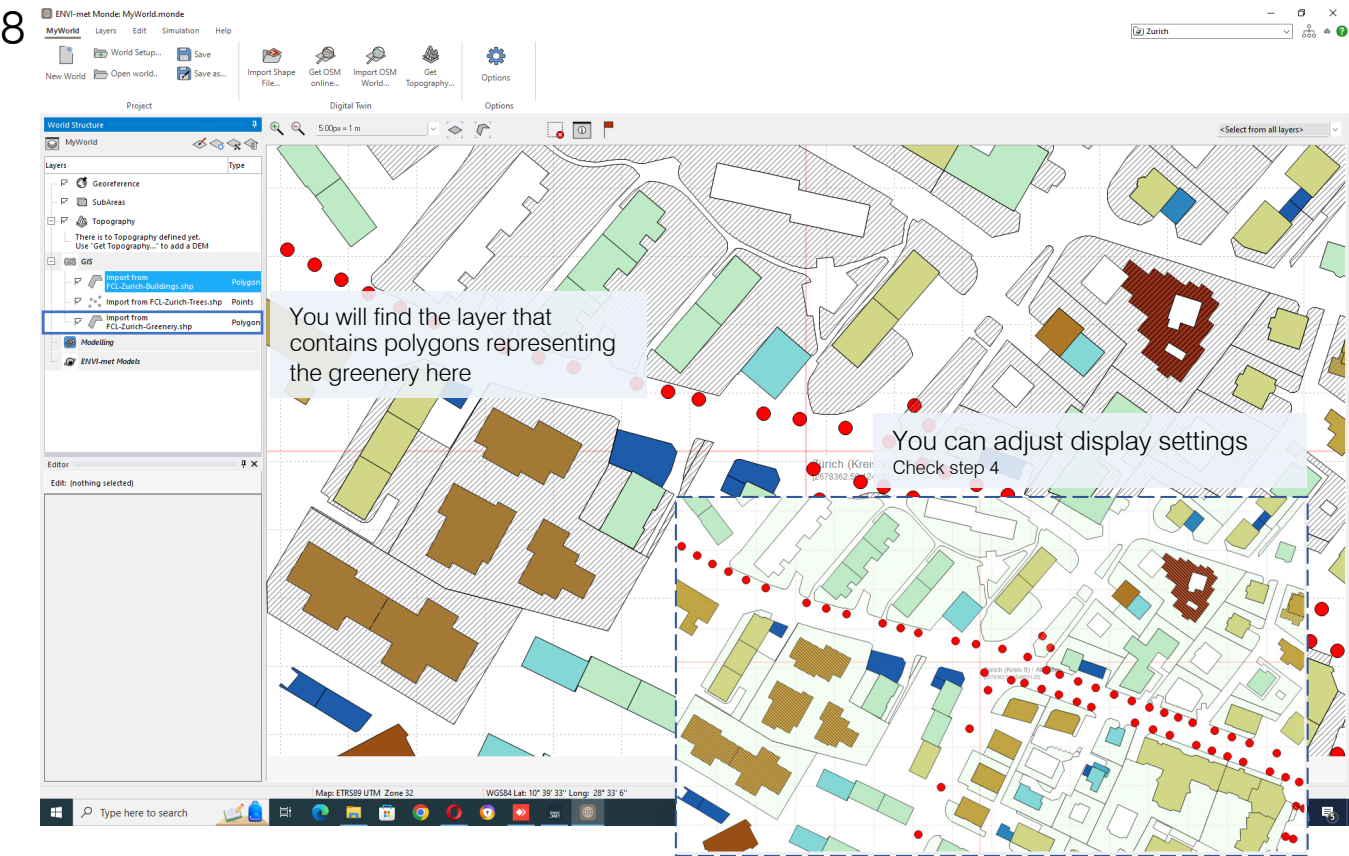
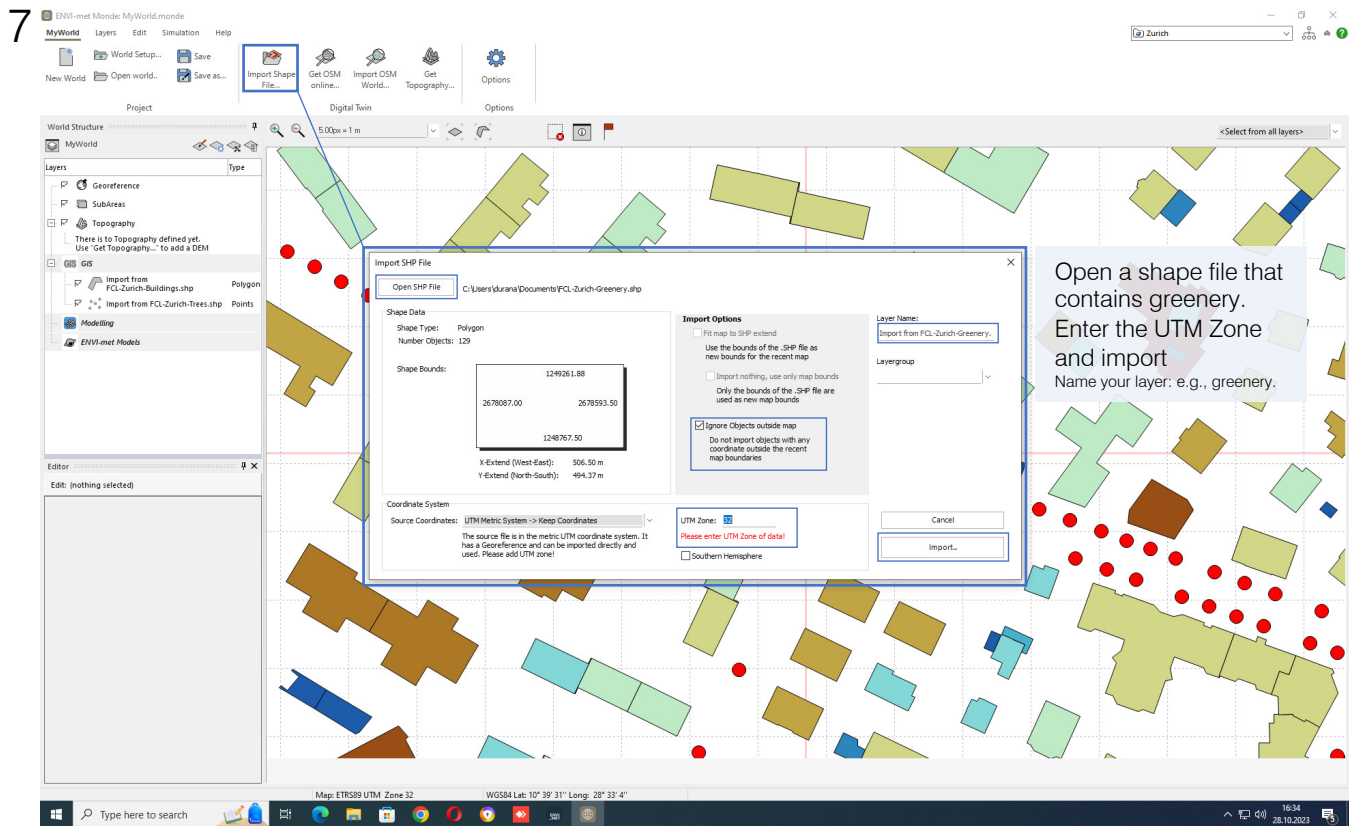
5



6

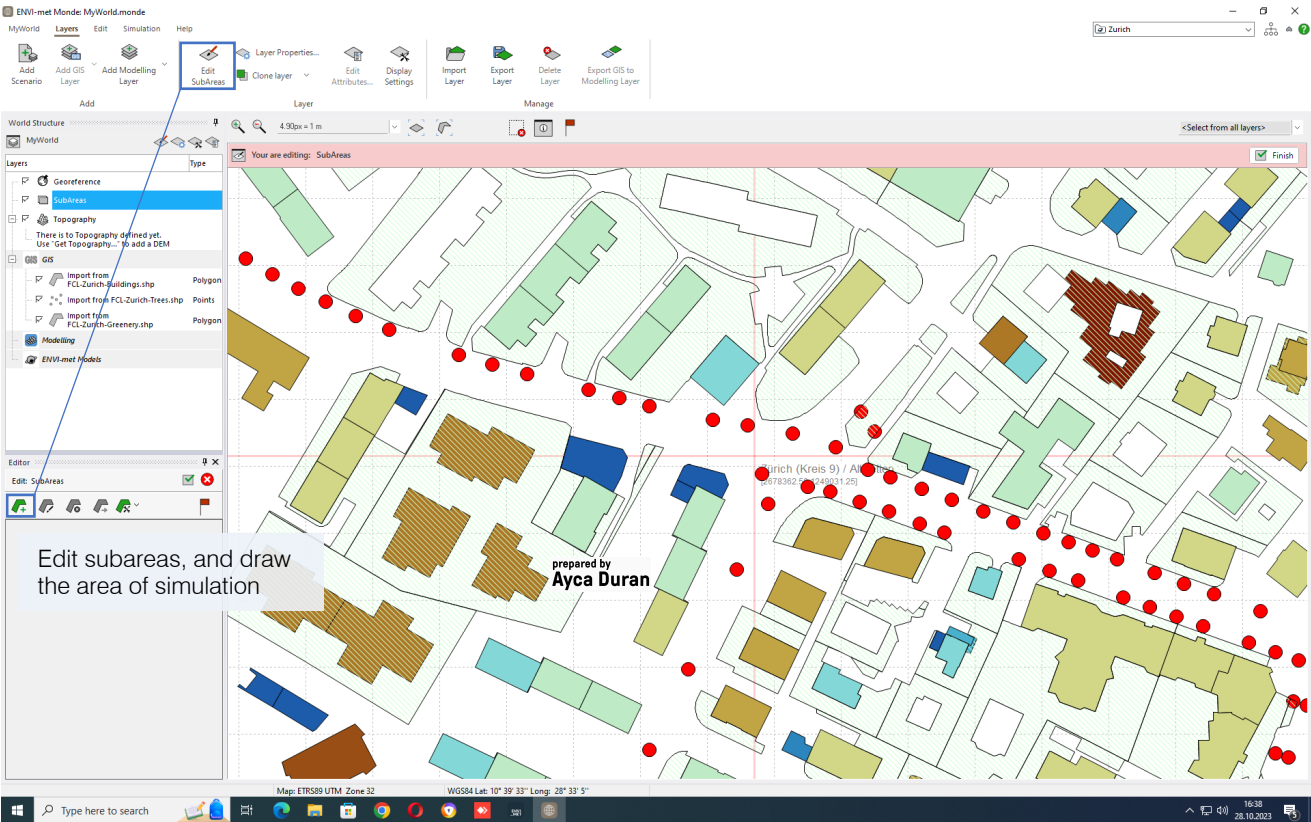


Modeling

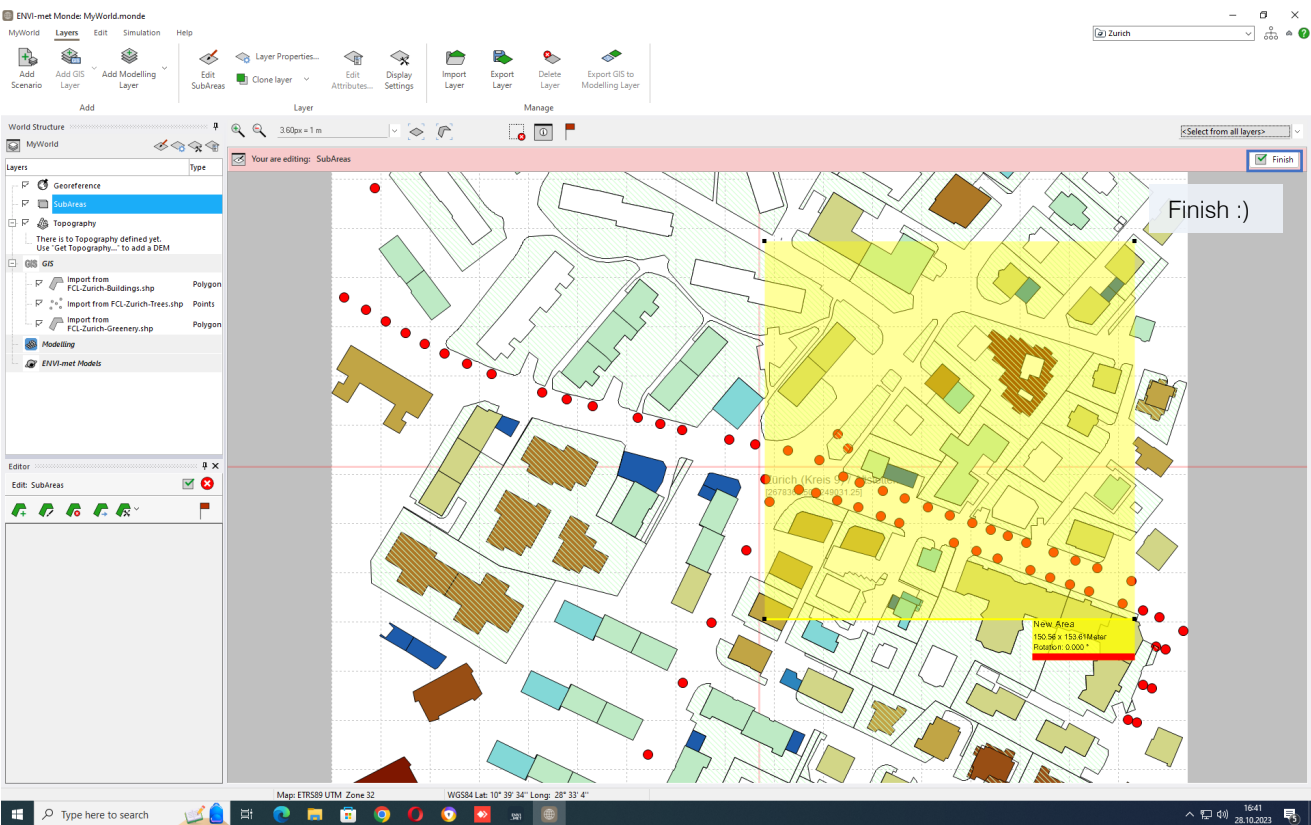


Modeling

9

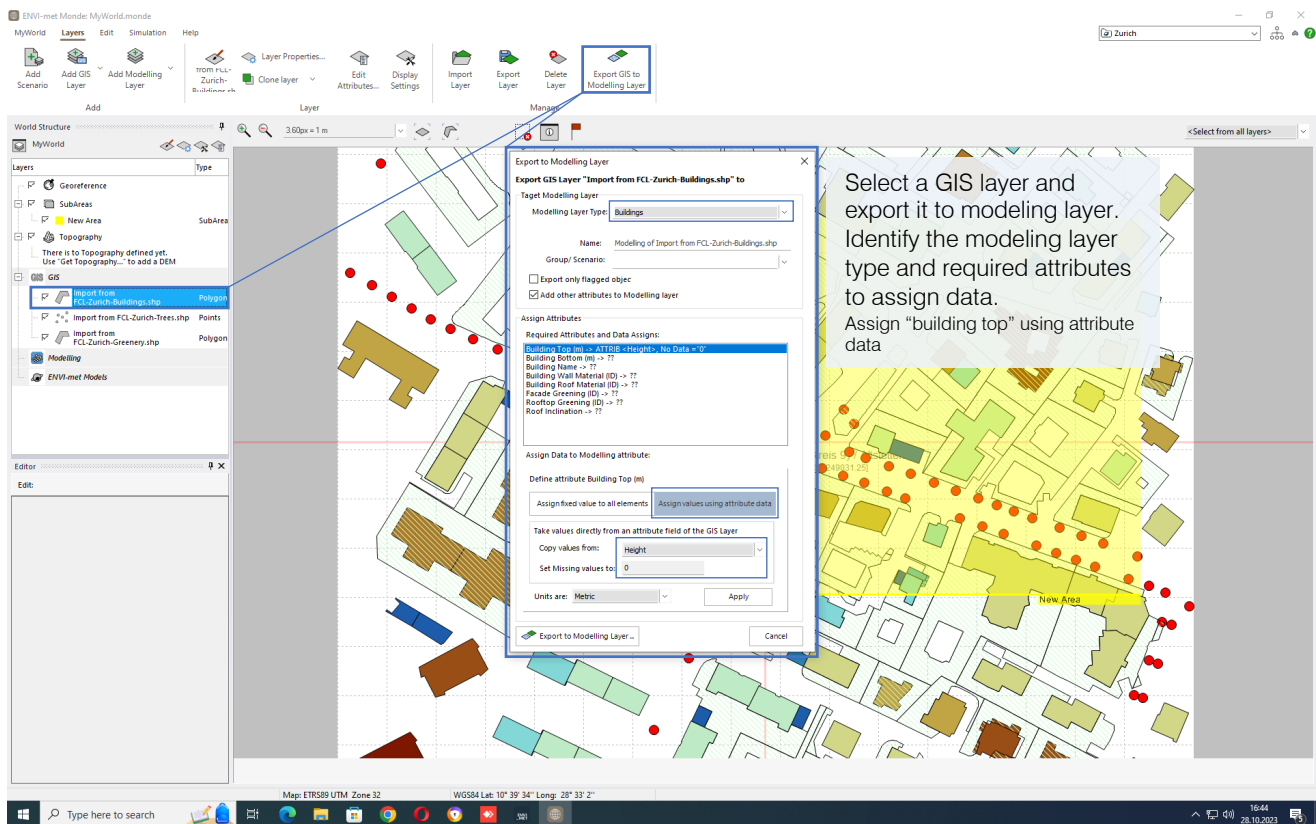


10

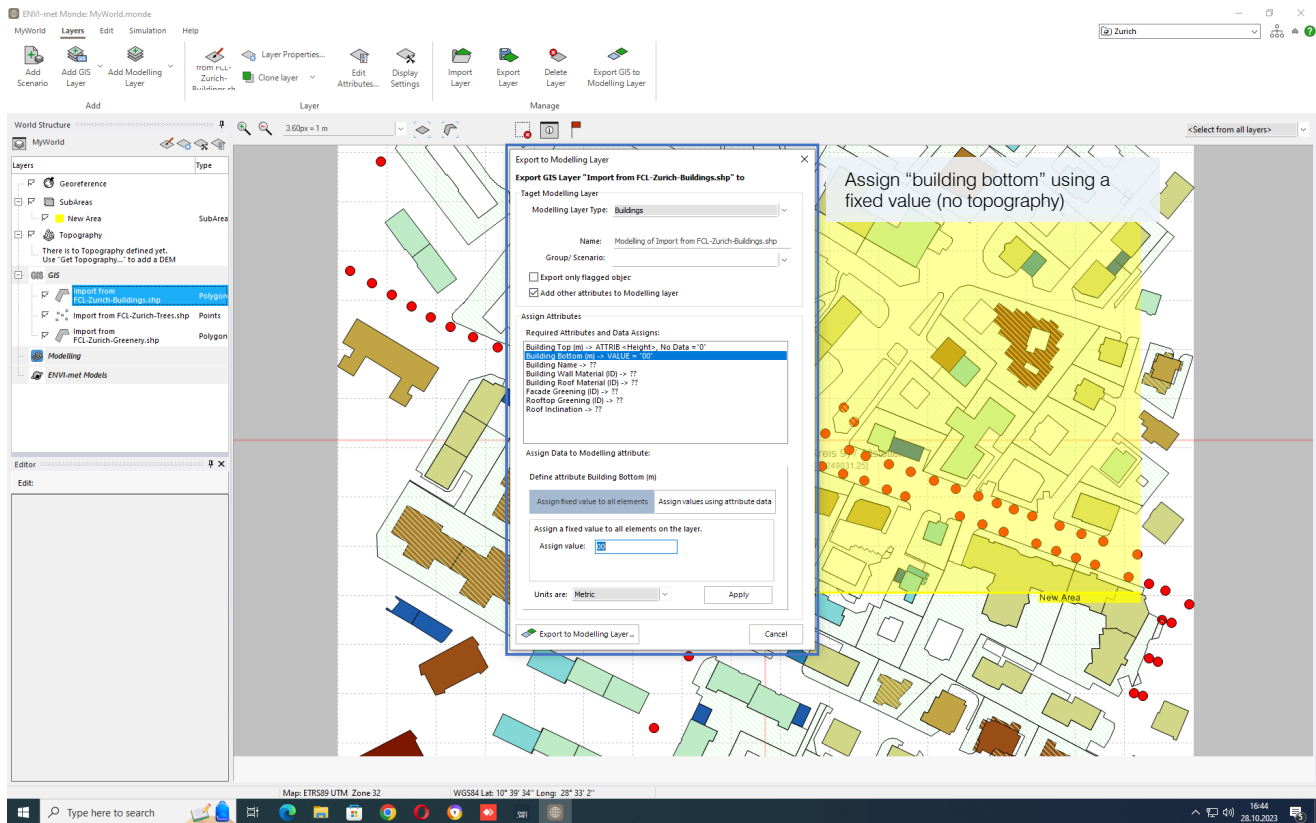


Modeling

11

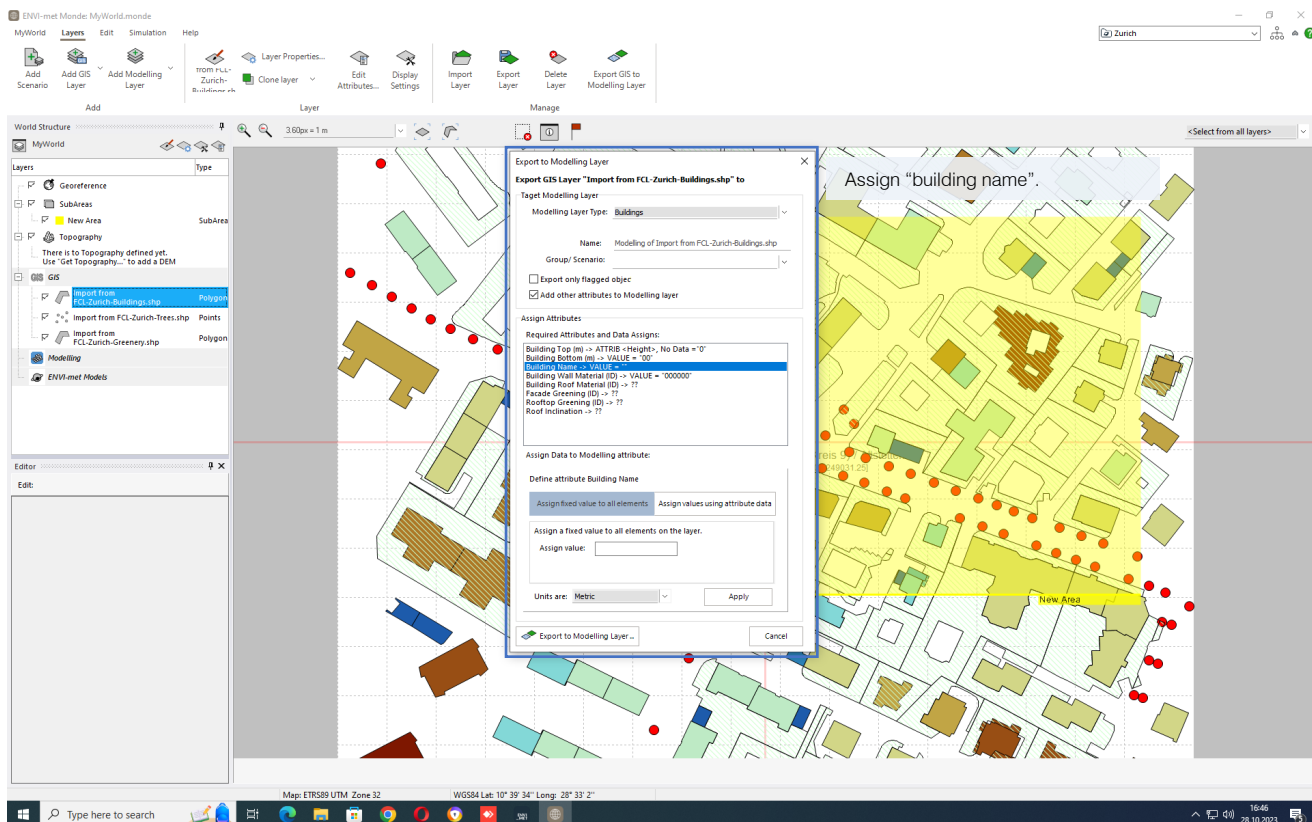


12

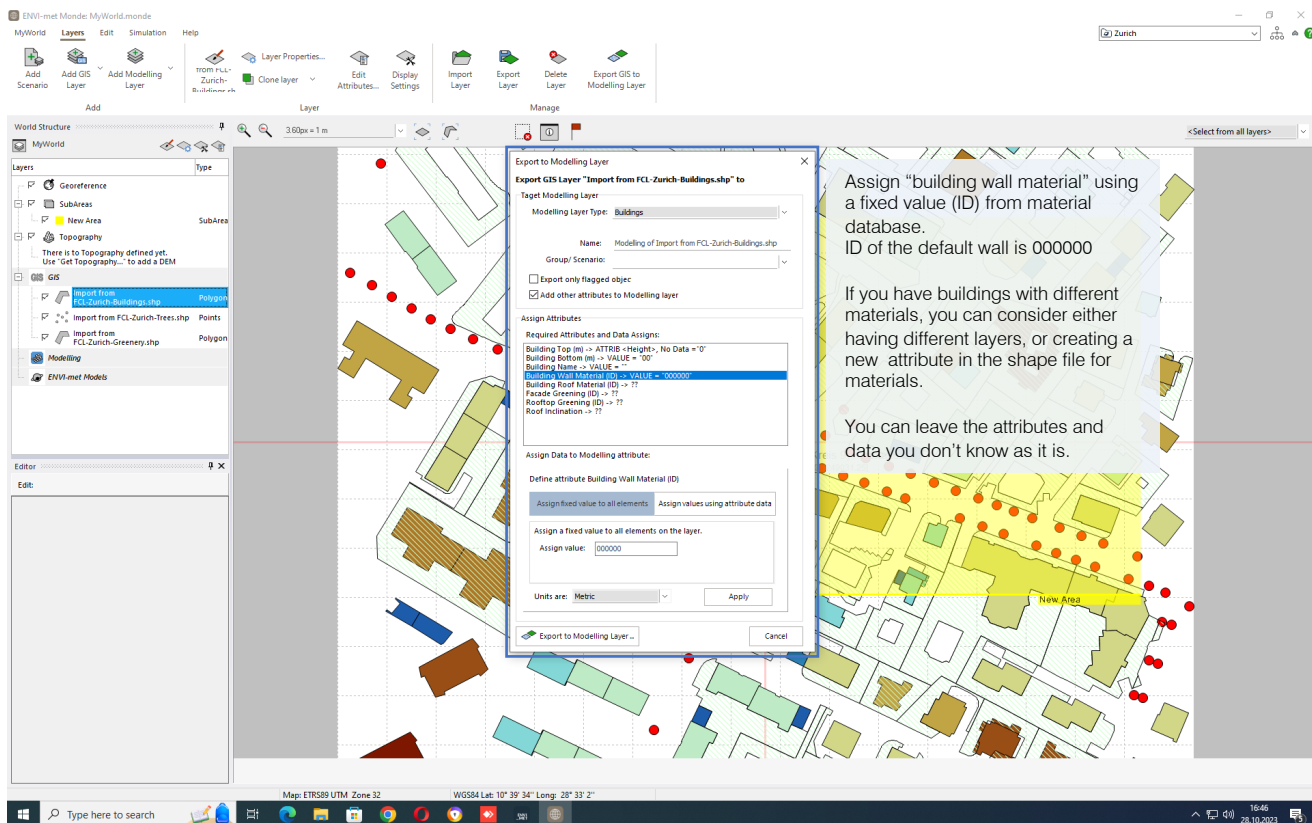


Modeling

13



14



Modeling

15

ENVI-met Monde: MyWorld.monde

World Structure: MyWorld

Layers: Georeference, SubAreas, New Area, Topography, GIS, Modeling, ENVI-met Models

Export to Modelling Layer

Export GIS Layer "Import from FCL-Zurich-Buildings.shp" to

Target Modelling Layer: Buildings

Modelling Layer Type: Buildings

Name: Modeling of Import from FCL-Zurich-Buildings.shp

Group/ Scenario:

☐ Export only flagged object

☒ Add other attributes to Modelling layer

Assign Attributes:

Required Attributes and Data Assigns:

Building Top (m) -> ATTRIBUTE "Height", No Data = 0"

Building Bottom (m) -> VALUE = "00"

Building Name -> VALUE =

Building Wall Material (ID) -> VALUE = "000000"

Building Roof Material (ID) -> VALUE = "000000"

Facade Greening (ID) -> ??

Rooftop Greening (ID) -> ??

Roof inclination -> ??

Assign Data to Modelling attribute:

Define attribute Building Roof Material (ID)

Assign fixed value to all elements

Assign values using attribute data

Assign a fixed value to all elements on the layer.

Assign value: 000000

Units are: Metric

Apply

Export to Modelling Layer

Cancel

Assign "building roof material" using a fixed value (ID) from material database.
ID of the default roof is 000000.

If you have buildings with different materials, you can consider either having different layers, or creating an attribute layer in the shape file for materials.

16

ENVI-met Monde: MyWorld.monde

World Structure: MyWorld

Layers: Georeference, SubAreas, New Area, Topography, GIS, Modeling, ENVI-met Models

Export to Modelling Layer

Export GIS Layer "Import from FCL-Zurich-Trees.shp" to

Target Modelling Layer: Vegetation

Modelling Layer Type: Vegetation

Name: Modeling of Import from FCL-Zurich-Trees.shp

Group/ Scenario:

☐ Export only flagged object

☒ Add other attributes to Modelling layer

Assign Attributes:

Required Attributes and Data Assigns:

3D Plant Type (ID) -> VALUE = "02OLDM"

Assign Data to Modelling attribute:

Define attribute 3D Plant Type (ID)

Assign fixed value to all elements

Assign values using attribute data

Assign a fixed value to all elements on the layer.

Assign value: 02OLDM

Units are: Metric

Apply

Export to Modelling Layer

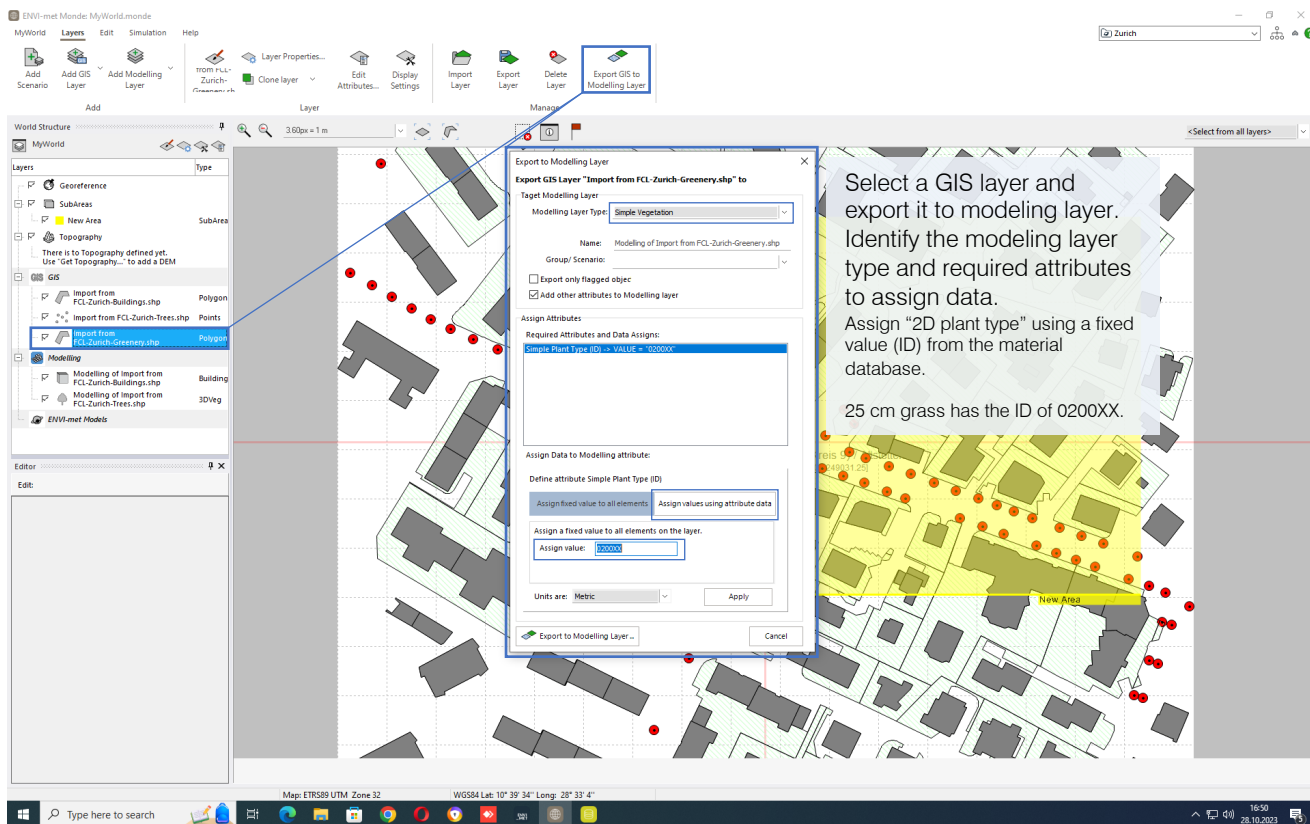
Cancel

Select a GIS layer and export it to modeling layer. Identify the modeling layer type and required attributes to assign data.

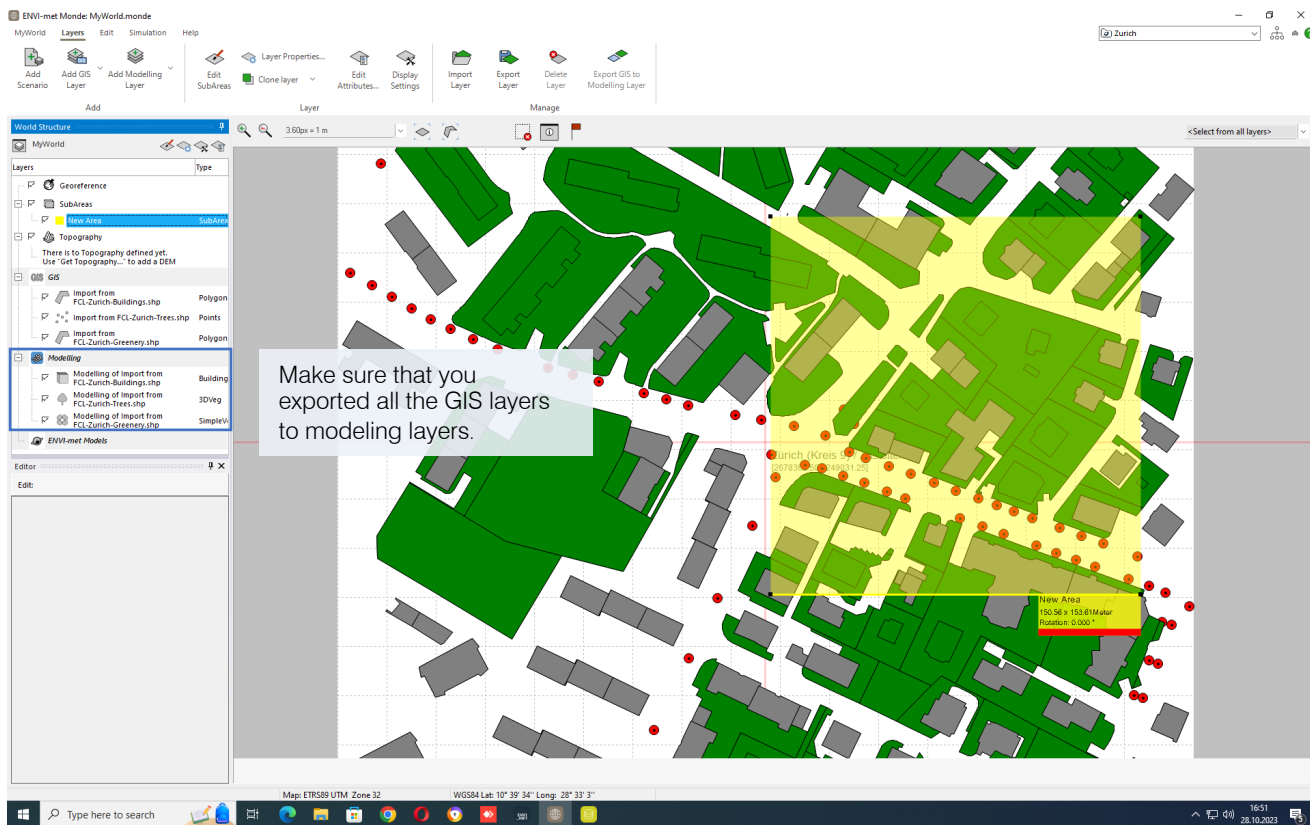
Assign "3D plant type" using a fixed value (ID) from the material database.

An abstract deciduous tree with medium canopy has the ID of 02OLDM.

17



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Modeling

19

ENVI-met Monde: Altstadt_zurich_demo.monde

MyWorld Layers Edit Simulation Help

Subarea New Area

Area Size: 150.56 x 153.61 Meter

Area Rotation: 0.00°

Create ENVI-met Model...

Export Model to INX File...

Analyse...

World Structure

MyWorld

Layers

Georeference

SubAreas

New Area

Topography

There is no Topography defined yet. Use 'Get Topography...' to add a DEM

GIS

Import from FCL-Zurich-Buildings.shp Polygon

Import from FCL-Zurich-Trees.shp Points

Import from FCL-Zurich-Greenery.shp Polygon

Modeling

Modeling of import from FCL-Zurich-Buildings.shp Building

Modeling of import from FCL-Zurich-Trees.shp 3Dveg

Modeling of import from FCL-Zurich-Greenery.shp SimpleVe

ENVI-met Models

Editor

Edit: (nothing selected)

Create ENVI-met Model

Select Data

Use Scenario: Only Layers not linked to Scenarios/Groups

Select Modeling Layers for Model:

Modeling Layer	Type
☒ Topography	Topography
☒ Buildings	Building Data (2,3D)
☒ Vegetation	3D Vegetation
☒ Surface	Simple Vegetation
☐ Sources	
☐ Receptors	

Layers not linked to a Scenario or Group are always available

Model Location

Name: Zurich (Viere 9) / Altstadt

Longitude: 28° 33' 2" Latitude: 10° 39' 24"

Time Zone: Unknown (0° 0' 0")

Model Domain

Build from Sub-Area: New Area

Area Size: 150.56 x 153.61 Meter

Area Rotation: 0.00°

Output Settings

Horizontal Resolution (m): 2.0

Resulting model size: 76 x 77 Grids

Vertical Resolution (m): 2

Amount of Z grids: 25

☐ dz of lowest gridbox is split into 5 subcells

☐ telescoping (dz increases with height)

Telescoping factor (%): 0

Start telescoping after height (m): 0

Model Name: Export Area New Area

Remark: Model Export from MyWorld

More Options...

Process

Create Model... Close

ENVI-MET

Decoding Urban Nature

Map: ETRS89 UTM Zone 32

WGS84 Lat: 10° 39' 34" Long: 28° 33' 5"

Type here to search

16:05

29.10.2023

Create the model when you're done.

20

ENVI-met Monde: MyWorld.monde

MyWorld Layers Edit Simulation Help

Subarea New Area

Area Size: 150.56 x 153.61 Meter

Area Rotation: 0.00°

Create ENVI-met Model...

Export Model to INX File...

Analyse...

World Structure

MyWorld

Layers

Georeference

SubAreas

New Area

Topography

There is no Topography defined yet. Use 'Get Topography...' to add a DEM

GIS

Import from FCL-Zurich-Buildings.shp Polygon

Import from FCL-Zurich-Trees.shp Points

Import from FCL-Zurich-Greenery.shp Polygon

Modeling

Modeling of import from FCL-Zurich-Buildings.shp Building

Modeling of import from FCL-Zurich-Trees.shp 3Dveg

Modeling of import from FCL-Zurich-Greenery.shp SimpleVe

ENVI-met Models

Editor

Edit:

Create ENVI-met Model

Select Data

Use Scenario: Only Layers not linked to Scenarios/Groups

Select Modeling Layers for Model:

Modeling Layer	Type
☒ Buildings	Building Data (2,3D)

Model Preview: Export Area New Area

Display Model Layer: Buildings

Show Model Grid

Model Preview

This is a preview of the gridded ENVI-met model generated by Monde.

If it fits the expectations, click 'Add to map' to include it in the recent map. If you want to change it, press 'Discard model'.

Export the model to an INX file

(To export the model later, use the context menu of the world structure overview)

Add to map... Discard model

ENVI-MET

Map: ETRS89 UTM Zone 32

WGS84 Lat: 10° 39' 34" Long: 28° 33' 3"

Type here to search

16:54

28.10.2023

Save your model to INX file.